

DETROIT ST CLAIR SHORES, MI, USA

WMO: 997996

Lat: 42.4711N

Lon: 82.8766W

Elev: 180

StdP: 99.18

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.6	-13.8	N/A	N/A	N/A	N/A	N/A	N/A	11.8	-4.5	10.8	-5.1	3.8	230	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	31.5	N/A	29.7	N/A	28.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.2	8.3	DB	-19.8	34.5	4.6	1.6	-23.1	35.7	-25.8	36.6	-28.4	37.5	-31.8	38.7	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-4.4	-3.6	1.3	8.6	15.3	20.4	23.4	22.2	18.7	12.0	5.1	0.1
	DBStd	10.68	5.96	5.60	5.30	4.31	4.41	3.63	3.26	2.70	3.75	3.95	4.78	4.52
	HDD10.0	1676	446	382	276	77	6	0	0	0	0	24	158	308
	HDD18.3	3483	703	615	529	293	116	19	2	3	41	199	397	566
	CDD10.0	1672	1	1	4	35	171	312	414	378	259	87	11	1
	CDD18.3	438	0	0	0	1	23	80	158	123	51	3	0	0
	CDH23.3	2593	0	0	0	16	144	461	1107	638	219	8	0	0
	CDH26.7	680	0	0	0	2	34	142	323	124	54	1	0	0
Wind	WSAvg	4.0	4.3	4.2	4.4	4.7	4.0	3.8	3.5	3.5	3.7	4.2	4.2	4.2
Precipitation	PrecAvg	890	59	54	63	83	92	85	89	80	86	66	67	68
	PrecMax	1352	113	134	141	146	177	154	198	148	183	155	164	121
	PrecMin	685	19	13	20	22	36	39	27	15	23	12	16	23
	PrecStd	138	25	28	30	32	41	33	42	37	47	30	31	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.5	13.4	18.0	26.1	30.0	33.1	34.9	31.8	32.0	24.3	18.6	14.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	10.4	9.5	15.0	21.4	27.1	30.3	31.8	29.7	27.8	21.1	15.7	11.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	6.3	6.2	11.8	18.5	24.6	28.1	30.0	28.0	25.6	19.7	13.9	8.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	3.3	3.1	8.7	15.3	22.1	25.8	28.3	26.6	24.1	17.9	11.9	6.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range		MDBR	6.4	6.7	7.2	7.8	7.9	7.0	7.2	6.7	6.9	6.7	5.9	5.2
		MCDBR	9.4	10.4	11.3	13.0	11.0	9.7	9.2	8.3	8.5	8.4	8.9	8.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.304	0.334	0.352	0.397	0.428	0.445	0.435	0.426	0.388	0.360	0.321	0.308	
	taud	2.388	2.288	2.352	2.246	2.213	2.202	2.255	2.269	2.356	2.418	2.490	2.425	
	Ebn at Noon	834	859	891	870	849	833	837	833	841	823	813	796	
	Edh at Noon	80	103	110	132	140	142	134	128	109	91	72	71	
All-Sky Solar Radiation	RadAvg	1.44	2.33	3.51	4.63	5.42	6.13	6.11	5.29	4.27	2.77	1.71	1.22	
	RadStd	0.16	0.25	0.35	0.41	0.46	0.38	0.30	0.21	0.37	0.24	0.22	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page